Validity Analysis: DRBENCHMARK

Target context: French Pharmaceutical Regulatory Affairs — Document Compliance NLP

Overall risk: **HIGH**

Deployment Context

Use case: In a hospital setting, a clinician uses software to manage unstructured French clinical cases by applying a model evaluated on part-of-speech tagging, named-entity recognition, and multi-label classification. The system identifies specific clinical and temporal entities while mapping pathologies to standardized medical classification chapters and axes to automatically generate a patient's clinical profile. These outputs directly help the clinician summarize complex medical histories and prioritize care based on the severity of identified symptoms, ensuring that life-critical decisions are supported by accurately synthesized data.

Target population: French-speaking medical professionals, including clinicians, hospital physicians, and nursing staff in French-speaking health systems.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ⚠	2	Significant gaps	high
Input Content	2	Significant gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology ⚠	1	Serious concern	high
Output Content ⚠	2	Significant gaps	high
Output Form ✓	3	Moderate gaps	medium
Average	**2.3**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

DrBenchmark is the most relevant existing French biomedical NLP benchmark for this deployment but is fundamentally misaligned with French pharmaceutical regulatory affairs compliance-checking. Strengths concentrate in input form (text-only standard French, HuggingFace toolkit compatibility, tokenizer-agnostic evaluation) and partial NER content (QUAERO/emea, MANTRAGSC/fr_emea drug-leaflet text, DEFT2021 SUBSTANCE/DOSAGE NER, PxCorpus posology entities). The most consequential gaps are in output ontology — NER label spaces (UMLS Semantic Groups, CLINENTITY, ICD-10 chapters, MeSH C axes) do not capture regulatory entity sub-types (INN-vs-brand, ATC, excipient, MedDRA), and the 0–5 STS scale assigns high similarity to legally significant qualifier and dose-threshold differences (DEFT2020-D2, DEFT2020-D18, CLISTER-D17, CLISTER-D18) that would trigger Type IA/IB/II variation submissions under EMA/ANSM rules — and in input ontology, where no task is designed around SmPC/CTD/ANSM-format compliance checking. Output content is compromised by the absence of regulatory affairs annotators across all 11 datasets and the use of clinical/research annotation norms as ground truth. Input content shows substantial off-domain dilution (encyclopedic, veterinary, non-French content) and zero deliberate coverage of French overseas territory or tropical pathology vocabulary identified by the user as the highest secondary priority. Output form is broadly compatible but lacks confidence calibration evaluation. The benchmark can support general French biomedical model selection and provide partial training signal for pharmaceutical NER, but cannot validly evaluate regulatory compliance decisions without substantial supplementation.

ID	Evidence
DEFT2020-D2	En conséquence, par mesure de précaution, il convient d'éviter d'allaiter pendant la durée du traitement. — Breastfeeding contraindication with precautionary qualifier; cible drops qualifier
DEFT2020-D18	scores=[5.0, 2.0, 4.0, 4.0, 5.0] — Precautionary qualifier dropped in cible; rated 4.0 — legally significant difference not captured
CLISTER-D17	Le recul est de 2 ans." / "Le suivie est de 4 ans et demi. — Different follow-up durations scored 4.0; same rubric would conflate regulatory dose differences
CLISTER-D18	L'AHG urinaire est à 10 mmol/l." / "L'AHG urinaire est à 20 mmol/l. — Twofold quantitative difference scored 4.0; regulatory dose doubling requires near-0

Practical Guidance

What This Benchmark Measures

DrBenchmark measures general French biomedical language understanding across NER, STS, classification, and MCQA tasks drawn from clinical cases, research literature, drug leaflets, clinical trial protocols, and pharmacy exams. Its strongest signal for this deployment is comparative French biomedical model performance (input_form score 4) — useful for model shortlisting (DrBERT-FS, DrBERT-CP, CamemBERT-bio variants per Q60–Q62). It provides partial training signal for pharmaceutical NER through QUAERO/emea, MANTRAGSC/fr_emea, DEFT2021/ner, and PxCorpus, and partial STS exposure through DEFT2020 drug-leaflet pairs and CLISTER.

Construct Depth

Construct depth is shallow for the regulatory compliance use case. The benchmark probes general semantic proximity (STS) and clinical/research entity recognition (NER), not regulatory equivalence or regulatory entity sub-type recognition. The 0–5 STS rubric does not distinguish legally operative differences (precautionary qualifier removal, dose threshold changes, route additions) from semantically proximate variants — confirmed empirically in DEFT2020-D2/D18 and CLISTER-D17/D18. NER schemas conflate INNs, brand names, and excipients under single labels (QUAERO-D13, DEFT2021-D10), preventing evaluation of regulatory sub-type discrimination. Output content lacks regulatory affairs annotation authority across all 11 datasets.

What Else You Need

Substantial supplementation is required: (1) a purpose-built French regulatory NER test set with INN-vs-brand, ATC, excipient, and MedDRA-aligned entity types, annotated by regulatory affairs specialists (addresses output_ontology and output_content gaps); (2) a regulatory-equivalence STS test set calibrated to Type IA/IB/II variation classifications with ground truth from regulatory experts (addresses output_ontology); (3) SmPC and PIL document samples for compliance-checking task evaluation (addresses input_ontology and input_content); (4) French overseas territory tropical pathology vocabulary supplementation if DOM deployment is pursued (addresses input_content); (5) confidence calibration and human-review threshold evaluation tied to ANSM workflow requirements (addresses output_form). Until these are constructed, DrBenchmark should be treated as a screening filter for model selection, not as a deployment-readiness gate.

ID	Evidence
Q60	“French biomedical language models (DrBERT-FS, DrBERT-CP, CamemBERT-bio), presumed to be the most aligned with the nature of the data of the benchmark, exhibit indeed superior performance across many tasks.” (p.6)
Q61	“More precisely, DrBERT-FS achieves the highest performance in 8 tasks, DrBERT-CP in 5 tasks, and CamemBERT-bio in 2 tasks.” (p.6)
Q62	“This indicates that domain and language-specialized models achieve the best performance in up to 75% of the DrBenchmark downstream tasks.” (p.6)
DEFT2020-D2	En conséquence, par mesure de précaution, il convient d'éviter d'allaiter pendant la durée du traitement. — Breastfeeding contraindication with precautionary qualifier; cible drops qualifier
CLISTER-D17	Le recul est de 2 ans." / "Le suivie est de 4 ans et demi. — Different follow-up durations scored 4.0; same rubric would conflate regulatory dose differences
QUAERO-D13	Refludan 50 mg en poudre pour solution injectable ... Lepirudine Les autres composants sont le mannitol (E 421) et l' hydroxyde de sodium — Brand name, INN, excipients all share CHEM label — regulatory distinction collapsed
DEFT2021-D10	diphenhydramine et de méthylprednisolone — Drug names tagged SUBSTANCE with no INN/ATC/excipient distinction